WEEK-2 Practical

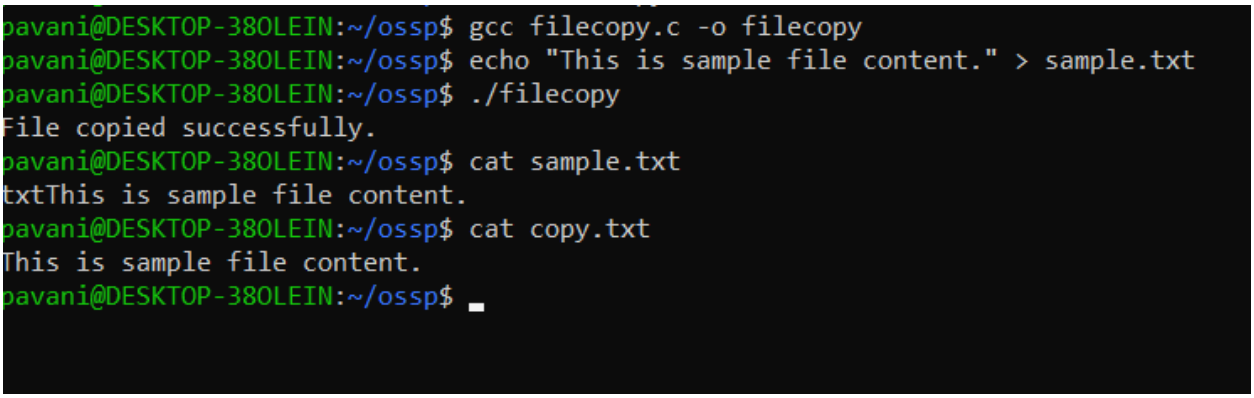
Task-1 Develop a C program that uses the system calls open(), read(), write(), and closeo to copy the contents of one file to 3 another. Explain how control transitions between user Space and kernel space during execution.

pavani@DESKTOP-38OLEIN: ~/ossp

GNU nano 8.7.1 filecopy.c *

```c
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <stdlib.h>

int main() {
    int source, destination;
    char buffer[1024];
    ssize_t bytesRead, bytesWritten;

    source = open("sample.txt", O_RDONLY);

    if (source == -1) {
        perror("Error opening source file");
        return 1;
    }

    destination = open("copy.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (destination == -1) {
        perror("Error opening destination file");
        close(source);
        return 1;
    }

    while ((bytesRead = read(source, buffer, sizeof(buffer))) > 0) {

        bytesWritten = write(destination, buffer, bytesRead);

        if (bytesWritten != bytesRead) {
            perror("Error writing to destination file");
            close(source);
            close(destination);
            return 1;
        }
    }

    if (bytesRead == -1) {
        perror("Error reading source file");
    }

    close(source);
    close(destination);

    printf("File copied successfully.\n");

    return 0;
```

Output:



Report:

In this task, I developed a C program to copy the contents of one file into another file using the system calls open(), read(), write() and close(). The source file was opened in read mode and the destination file was opened in write mode. The contents were read into a buffer and written into the destination file. After completing the operation, both files were closed. I also studied how the control moves from user space to kernel space whenever a system call is made. The program was executed successfully and the file contents were copied correctly.

Task-2

Use the strace utility to trace all system calls generated by the command cat sample.txt. Analyze the sequence of system 3 calls and identify the kernel services involved

```
pavani@DESKTOP-38OLEIN:~/ossp$ strace -e trace=openat,read,write,close cat sample.txt
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
close(3)                                = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
close(3)                                = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libgcc_s.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
close(3)                                = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libm.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
close(3)                                = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0 \250\2\0\0\0\0\0"..., 832) = 832
close(3)                                = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libpcre2-8.so.0", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
close(3)                                = 0
openat(AT_FDCWD, "/proc/filesystems", O_RDONLY|O_CLOEXEC) = 3
read(3, "nodev\tsysfs\nnodev\ttmpfs\nnodev\tbd"..., 1024) = 442
close(3)                                = 0
openat(AT_FDCWD, "/proc/mounts", O_RDONLY|O_CLOEXEC) = 3
read(3, "none /usr/lib/modules/6.18.33.2-"..., 1024) = 1024
read(3, "e 0 0\ndevpts /dev/pts devpts rw,"..., 1024) = 1024
read(3, "osuid,nodev,noexec,relatime,nosy"..., 1024) = 1024
read(3, "relatime 0 0\nnone /run/credentia"..., 1024) = 383
read(3, "", 1024)                       = 0
close(3)                                = 0
openat(AT_FDCWD, "/proc/self/maps", O_RDONLY|O_CLOEXEC) = 3
read(3, "5b62fa897000-5b62fa989000 r--p 0"..., 1024) = 1024
read(3, "93fe000-7c9f093ff000 r--p 000a90"..., 1024) = 1024
read(3, "00 08:30 13631                  "..., 1024) = 1024
read(3, "             /usr/lib/x86_64-lin"..., 1024) = 1024
read(3, "0 13625                      /us"..., 1024) = 566
close(3)                                = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/uucore/en-US.ftl", O_RDONLY|O_CLOEXEC) = 3
read(3, "# Common strings shared across a"..., 4080) = 4080
read(3, "", 32)                         = 0
close(3)                                = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/cat/en-US.ftl", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such
file or directory)
openat(AT_FDCWD, "sample.txt", O_RDONLY|O_CLOEXEC) = 3
This is sample file content.
close(5)                                = 0
close(4)                                = 0
close(3)                                = 0
+++ exited with 0 +++
pavani@DESKTOP-38OLEIN:~/ossp$
```

Report: In this task, I used the strace utility to trace the system calls generated by the command cat sample.txt. The important system calls observed were openat(), read(), write() and close(). The openat() call was used to open the file, read() was used to read its contents, and write() displayed the contents on the terminal. Finally, close() released the file descriptor. From the trace, I understood how the cat command interacts with the kernel through system calls.